# Final Evaluation – DAA Mock Interview

**Student Name:** Shaik Reehan
**Register Number:** 192565071
**Course:** Design and Analysis of Algorithms
**Evaluation:** CO5 – AT3 DAA Mock Interview
**Total Marks: 100**

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## 1. Question-Wise Performance

| Q.No. | Topic | Performance | Level |
|---|---|---|---|
| 1 | Backtracking | Clearly explained backtracking, pruning, and brute-force difference with examples. | Excellent |
| 2 | State-Space Tree | Correctly explained nodes, branches, and pruning. | Excellent |
| 3 | N-Queens | Correctly identified column and diagonal constraints. | Excellent |
| 4 | Subset Sum | Correctly explained include/exclude branching and pruning. | Excellent |
| 5 | Graph Coloring | Correctly explained color constraints and when to backtrack. | Very Good |
| 6 | P, NP, NP-Hard, NP-Complete | Correctly explained the four categories and their relationships. | Very Good |
| 7 | Polynomial-Time Reduction | Correctly explained transformation and its importance in NP-Completeness. | Very Good |
| 8 | Master Theorem | Correctly calculated $a = 2$, $b = 2$, and obtained $\Theta(n \log n)$. | Excellent |
| 9 | Greedy vs DP | Correctly distinguished both techniques and explained Fractional Knapsack. | Excellent |
| 10 | Hamiltonian Cycle | Correctly explained the definition and backtracking constraints. | Excellent |

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# 2. Technical Strengths

The student demonstrated the following strengths:

- Strong understanding of **Backtracking**.
- Correct understanding of **pruning and constraint checking**.
- Good knowledge of **State-Space Trees**.
- Correct identification of constraints in **N-Queens**.

- Good understanding of **Subset Sum** using include/exclude decisions.
- Correct approach to **Graph Coloring**.
- Good understanding of **P, NP, NP-Hard, and NP-Complete**.
- Correct explanation of **Polynomial-Time Reduction**.
- Excellent application of the **Master Theorem**.
- Good distinction between **Greedy and Dynamic Programming**.
- Correct understanding of **Hamiltonian Cycle** and its backtracking approach.
- Able to explain concepts using practical examples.

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# 3. Areas Requiring Improvement

### 1. Formal Algorithmic Terminology

The student should practice giving answers using formal DAA terminology rather than relying mainly on analogies.

### 2. Complexity Analysis

For algorithm questions, the student should consistently state:

- Time complexity
- Space complexity
- Best/average/worst case when relevant

### 3. NP-Complete Concepts

The definitions of P, NP, NP-Hard, and NP-Complete should be practiced using precise mathematical terminology.

### 4. Graph Algorithms

The student should further practice:

- Hamiltonian Cycle
- Graph Coloring
- Edge Coloring
- Graph traversal
- Complexity analysis of graph algorithms

### 5. Concise Viva Answers

Some answers could be made more concise. A good structure for Viva answers is:

**Definition → Algorithm/Approach → Example → Complexity**

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# 4. Rubric Criteria and Level Achieved

Since the interview performance was evaluated across the demonstrated DAA concepts, the following rubric can be used for the final evaluation.

| Criterion | Level Achieved | Marks |
|---|---|---|
| **1. Fundamental Algorithmic Concepts** | Excellent | 15/15 |
| **2. Backtracking & State-Space Understanding** | Excellent | 15/15 |
| **3. Problem-Solving & Algorithm Application** | Excellent | 15/15 |
| **4. Complexity Analysis** | Very Good | 13/15 |
| **5. Graph Algorithms & Constraints** | Very Good | 13/15 |
| **6. Computational Complexity (P/NP)** | Very Good | 12/15 |
| **7. Dynamic Programming & Greedy Techniques** | Excellent | 10/10 |
| **8. Viva Communication & Technical Explanation** | Very Good | 10/10 |
| **Total** | **Very Good / Excellent** | **88/100** |

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# 5. Marks Obtained

### Detailed Breakdown

**Fundamental Algorithmic Concepts:**
15 / 15

**Backtracking & State-Space Understanding:**
15 / 15

**Problem-Solving & Algorithm Application:**
15 / 15

**Complexity Analysis:**
13 / 15

**Graph Algorithms & Constraints:**
13 / 15

**Computational Complexity:**
12 / 15

**Dynamic Programming & Greedy Techniques:**
10 / 10

**Viva Communication & Technical Explanation:**
10 / 10

### Total

$$15 + 15 + 15 + 13 + 13 + 12 + 10 + 10$$
$$\boxed{88/100}$$

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# 6. Overall Performance

### Overall Score: 88/100

### Overall Level: Very Good / Excellent

The student demonstrated a strong conceptual foundation in Design and Analysis of Algorithms. The performance was particularly strong in **backtracking, pruning, recurrence analysis, and algorithmic problem-solving**.

The student correctly solved the Master Theorem question and provided appropriate explanations for the major backtracking and graph problems.

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# 7. Specific Suggestions for Improvement

1. **Practice complexity analysis daily.**
   For every algorithm studied, identify its time and space complexity.
2. **Strengthen graph algorithm knowledge.**
   Practice N-Queens, Graph Coloring, Hamiltonian Cycle, BFS, DFS, and shortest-path problems.
3. **Revise NP-Completeness formally.**
   Memorize the relationships:

$$P \subseteq NP$$

   and

$$NP\text{-Complete} = NP \cap NP\text{-Hard}$$

4. **Practice recurrence problems.**
   Solve examples using:
   - Substitution Method
   - Recursion Tree
   - Master Theorem
5. **Improve Viva structure.**
   Answer questions using:

   **Definition → Working → Example → Complexity**

6. **Avoid over-explaining simple questions.**
   Give direct answers first and then provide an example if needed.
7. **Practice coding along with theory.**
   Be prepared to write or explain pseudocode for backtracking and dynamic-programming problems.

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# 8. Final Evaluator Remarks

**Shaik Reehan demonstrated very good knowledge of Design and Analysis of Algorithms during the final mock interview. The student showed strong understanding of backtracking, pruning, state-space trees, graph constraints, recurrence analysis, computational complexity, Greedy algorithms, and Dynamic Programming. The Master Theorem problem was solved correctly. Further improvement in formal complexity analysis, graph algorithms, and concise technical communication will help the student achieve an excellent level in the final Viva Voce.**

### Final Result

**Marks Obtained: 88 / 100**
**Performance Level: Very Good / Excellent**
**Recommendation: Continue practicing complexity analysis and graph/NP-complete problems for stronger final Viva performance.**